



Cloud Deployment & Disaster Recovery Guide

DEVOPS & INFRASTRUCTURE

Vercel Edge, Supabase Pooling & Migration Runbook

Effective Date: September 16, 2026
System: Order Control System v1.0

Cloud Deployment, DevOps & Disaster Recovery Guide

Order Control System — Multi-Brand Dark Kitchen Operating Platform

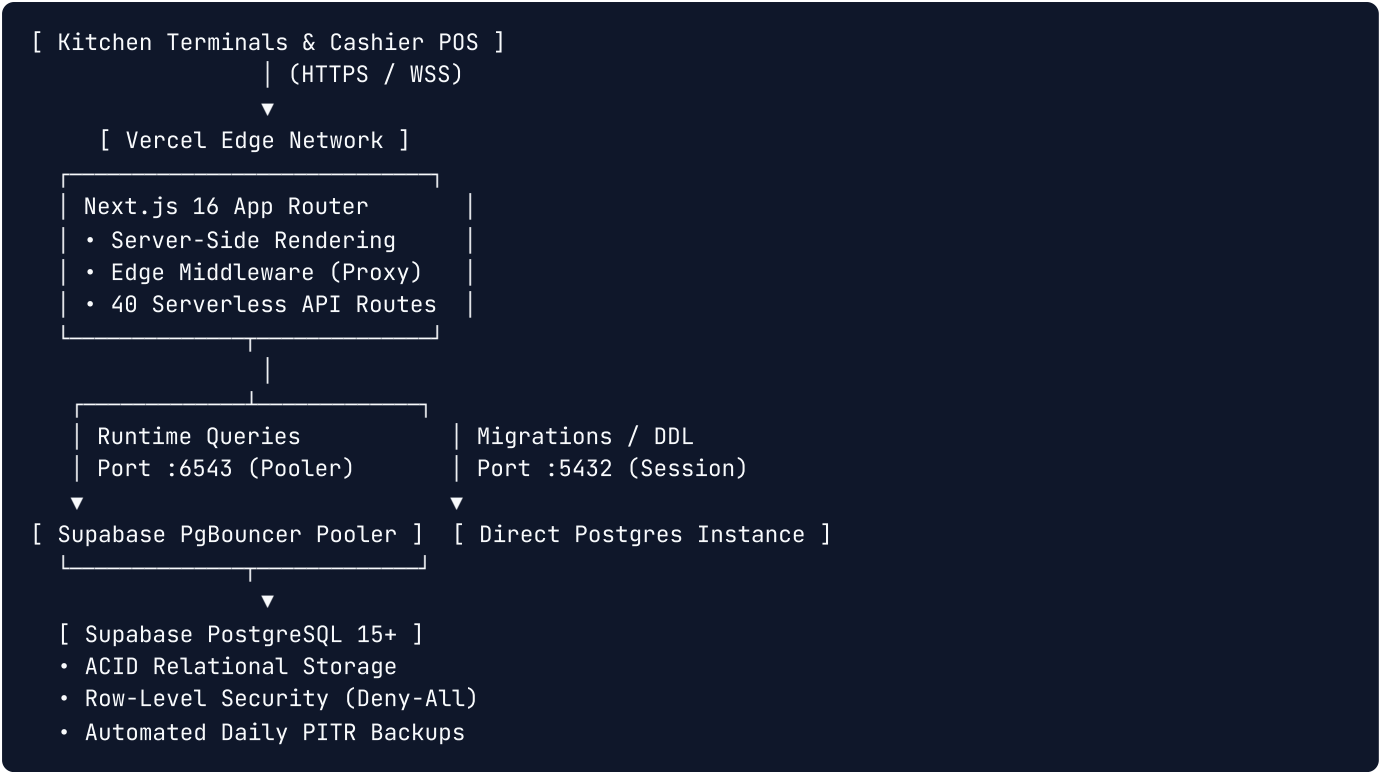
Printable Edition: [Download Executive PDF \(A4 Edition\)](#)
Deployment Target: Vercel (Edge & Serverless) + Supabase (PostgreSQL 15+)
Status: Production-Ready | **Revision:** September 2026

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1. Cloud Topology & Infrastructure Overview

The Order Control System is architected for zero-maintenance, serverless elasticity. It decouples edge presentation from relational persistence:



2. Environment Variables & Secrets Specification

Production deployments require six core environment variables configured in the Vercel Project Settings:

Environment Variable	Target / Scope	Format & Example	Operational Purpose
DATABASE_URL	Server Only	postgresql://postgres.[ref]:[pwd]@[host]:6543/postgres?pgbouncer=true	Runtime transactional queries via PgBouncer Pooler
DIRECT_URL	Build / Server	postgresql://postgres.[ref]:[pwd]@[host]:5432/postgres	Dedicated session connection for Prisma Migrations
NEXT_PUBLIC_SUPABASE_URL	Universal	https://[project-ref].supabase.co	Supabase API endpoint for browser & server auth
NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY	Browser	eyJhbGciOi...	Anonymous public key for client-side session renewal
SUPABASE_SERVICE_ROLE_KEY	Server Only	eyJhbGciOi...	Administrative key for user provisioning & management
OWNER_EMAIL	Server Only	owner@sushi.local	Designates initial administrator during bootstrap

 Critical Security Directives:

- Never expose SUPABASE_SERVICE_ROLE_KEY or DIRECT_URL to browser clients.
- Ensure .env is listed in .gitignore to prevent secret leaks to GitHub.

3. Database Migration Pipeline

Because the runtime connection DATABASE_URL uses PgBouncer in transaction mode (:6543), schema migrations and table alterations cannot run through the pooler. Prisma must use DIRECT_URL (:5432).

Automated CI/CD Migration Execution

During Vercel production builds or CI deployment pipelines, execute:

```
# 1. Apply all pending migrations safely to production
npx prisma migrate deploy

# 2. Generate updated Prisma 7 client bindings
npx prisma generate

# 3. Build Next.js application with Turbopack
npm run build
```

Creating New Schema Migrations Locally

When altering `prisma/schema.prisma` during development:

```
# Creates migration SQL file and applies it to dev database
npx prisma migrate dev --name <descriptive_name>
```

4. High-Concurrency Connection Pooling & Tuning

During kitchen rush hours (6:00 PM – 11:00 PM), dozens of orders may arrive concurrently. To eliminate pool exhaustion and database timeouts, the system enforces the following parameters in `src/lib/prisma.ts`:

```
// Interactive Transaction Timeout Configuration
export const TRANSACTION_OPTIONS = {
  maxWait: 15000, // Maximum time waiting for pool slot (15s)
  timeout: 25000, // Maximum transaction lifetime before rollback (25s)
};
```

Supabase Connection Pool Settings (Dashboard > Settings > Database)

- **Pool Mode:** `Transaction` (Mandatory for serverless scaling).
- **Default Pool Size:** `15 - 20` connections.
- **Max Client Connections:** `100+`.

5. Disaster Recovery & Backup Protocols (DR)

Strategy 1: Automated Point-In-Time Recovery (PITR)

Supabase Pro tiers execute continuous WAL archiving allowing second-by-second rollbacks up to 7 days:

- **Recovery Point Objective (RPO):** < 1 minute of data.
- **Recovery Time Objective (RTO):** < 15 minutes.
- **Trigger:** Initiated via Supabase Dashboard (`Database > Backups > Restore`).

Strategy 2: Manual Cold Snapshot (Pre-Deployment Safeguard)

Before executing high-risk schema migrations or large menu re-categorizations, take a manual database dump:

```
# Create timestamped SQL snapshot
pg_dump -d "$DIRECT_URL" -F c -b -v -f "backup_pre_deploy_$(date +%Y%m%d_%H%M%S).dump"

# Restore command in case of catastrophic migration failure
pg_restore -d "$DIRECT_URL" -v --clean "backup_pre_deploy_20260916_180000.dump"
```

6. First-Time Owner Provisioning Bootstrap

When launching a clean production instance:

1. Set the `OWNER_EMAIL` variable in Vercel to the owner's primary email address (e.g. `owner@sushi.local`).
2. Navigate to the `/login` page and sign in or execute initial signup.
3. The authentication hook in `src/lib/auth.ts` intercepts the login: if the user matches `OWNER_EMAIL` and no active owner exists, it automatically elevates the role to `OWNER`.
4. Subsequent staff accounts (Managers, Cashiers) are provisioned directly by the Owner from the `/users` dashboard.

7. Observability, Health Checks & Runbooks

Automated Health Verification

Deployments can be continuously monitored by pinging the lookups endpoint:

- **Health Endpoint:** `GET /api/lookups`
- **Expected Status:** `200 OK`
- **Payload Validation:** Returns active `brands` and `platforms`. A failure indicates database pool exhaustion or credentials expiration.

Operational Incident Checklist

Symptom	Root Cause	Remediation Procedure
P2024: Timed out fetching a new connection	PgBouncer pool saturated	Increase Supabase Pool Size or adjust serverless function concurrency limits.
P2028: Transaction API error: Transaction already closed	Long-running query exceeded 25s timeout	Optimize query indexes or split multi-table reporting aggregations into batches.
401 Unauthorized across all pages	Stale or invalidated Supabase Auth cookie	User must sign out and clear cookies, or verify Supabase project API keys in Vercel.
Duplicate key error on orderNumber	PostgreSQL Advisory Lock bypassed	Ensure all order creations invoke <code>src/services/orders.ts</code> and never bypass the service layer.

Official Production Deployment Runbook for Order Control System. Maintained by Infrastructure & DevOps Engineering.

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